ALTO — Personal Finance Tracker

PROJECT SOLUTION & SYSTEM DESIGN SPECIFICATION

■ **Live Deployed Application:** https://finance-g9aj.onrender.com	■ **GitHub Source Repository:** https://github.com/haricharananubolu/Finance
■ **Demo Account:** demo@alto.app	■ **Demo Password:** alto1234 (Pre-seeded with 22 transactions & 7 budgets)

1. Executive Summary & Problem Statement

The Problem: Individuals make poor financial choices because traditional tools provide passive post-mortem data rather than proactive, real-time threshold warnings before critical budget overspending occurs. Modern users managing high-velocity digital payments (UPI, Cards, Net Banking) struggle with unmonitored cash leaks and lack actionable clarity into savings velocity.

The Solution: *ALTO* is a full-stack personal finance management web application built with Python (Flask), SQLAlchemy ORM, SQLite, and Vanilla JavaScript. It delivers real-time net cash flow calculations, automated category budget health tracking (with predictive 80% caution and 100% cap breach alerts), interactive Chart.js visualizations, and strict per-user database isolation without intrusive third-party data tracking.

2. System Architecture & Technology Stack

Layer	Technology	Role & Technical Highlights
Backend	Python 3.12, Flask 3.0	Application Factory pattern with modular Blueprints (auth, dashboard, transactions, budgets, reports).
Database	SQLite, SQLAlchemy 2.0	ACID relational schema with fixed-point Decimal representation to eliminate monetary rounding errors.
Security	Flask-Login, Flask-WTF	Cryptographically secure Scrypt password hashing, CSRF tokens on all POST endpoints, and strict query isolation.
Frontend	HTML5, CSS3, Vanilla JS	Custom SaaS design system, responsive mobile drawer, and Chart.js 6-month comparative cash flow bar & doughnut charts.
Testing	Pytest 8.2 (19 Tests)	100% automated test coverage across authentication, CRUD workflows, budget health thresholds, and security isolation.
Hosting	Render Cloud, Gunicorn	Production WSGI deployment with zero-downtime continuous deployment from GitHub main branch.

3. Core Mathematical Algorithms & Budget Alert Engine

- **All-Time Net Balance:** Computed dynamically via $\sum$(Recorded Incomes) – $\sum$(Recorded Expenses).
- **Monthly Savings Rate (%):** ((Income - Expenses) / Income) × 100 when Income > 0, tracking wealth accumulation pace.
- **Automated Budget Thresholds:** Category limits automatically evaluate consumption against live expense entries:
  - **Healthy Status (< 80% limit):** Normal spending trajectory with teal indicators.
  - **Caution Status (80% – 100% limit):** Triggers amber warning banner before financial breach.
  - **Exceeded Status (> 100% limit):** Triggers high-priority rose alert banner indicating overspending.

4. Key Functional Modules

- **Real-Time Financial Dashboard:** Instant KPI cards, 6-month cash flow trends, category doughnut chart, and recent transaction feeds.
- **Transaction Manager:** Full CRUD workflows with search, multi-criteria filtering, sorting, pagination, and dynamic category dropdowns.
- **Monthly Budget Planner:** Target limit assignment per category with real-time visual progress meters reflecting actual expenses.
- **Reports & CSV Export:** Flexible period presets (This Month, Last Month, This Year, All Time) with one-click RFC-4180 CSV spreadsheet export.
- **User Preferences:** Multi-currency support (default ■ INR, $, €, £, ¥, AED) and secure in-app password changes.

5. Live Deployment & Evaluation Access

The application is fully operational in production. Evaluators can access the live service directly at **https://finance-g9aj.onrender.com** using the demo credentials provided above or by creating a new account via the registration portal.